

Project

Due: October 5, 2026

1 Speed

Your project is based on Scholtus and Dijk (2012), High-Frequency Technical Trading: The Importance of Speed. The paper is posted on bcourses. The paper looks at the impact of trading signals and the importance of the speed of execution. To do this, we will generate 3 trading signals and look at their effectiveness on three types of securities given instantaneous execution and delayed execution.

To do this, please:

- a. Choose 2 equities (one liquid and one less liquid), and one FX pair
- b. Choose a one-month period (e.g., Jan 2010)
- c. Choose two trading signals from the paper (e.g., MA, OBV) PLUS make your own trading signal
- d. Assume execution delays of 0ms, 100ms, 1 second
- e. Calculate the total returns on \$1,000,000 initial book value based on the signal + delay strategy
- f. Calculate the cost of delay (Figure 6 in the paper) for 100ms and 1 second execution delays

In all, there will be 18 profit-and-loss reports for each security/signal/delay combination. Please also comment on interesting effects you see. Some examples could be:

- a. Does one particular asset get affected more/less by execution delays? Why do you think that is?
- b. Do any of the chosen signals you use perform differently under certain market conditions?
- c. Does the tick frequency (the frequency with which you observe events, e.g., trades, quotes, orders) in each market affect the importance of execution delays? How would you factor that into your trading strategy?

2 RIT Simulation

Provide a write-up for the RIT ALGO2 Algorithmic Trading and ALGO2e done in the Discussion Session. This will be a different exercise. The first simulation (ALGO2) will be

to get used to use Market Orders and the *feebate* structure. This part will be individual, so every team member should submit a write-up. The second simulation (ALGO2e) will be done in Groups (We'll assign you a group name), so make sure that the group is together during the Session.

2.1 ALGO 2 - Individual

- a. What was your strategy going in?
- b. How would you decide what the appropriate bid/ask spread should be when you submit your pair orders? Should this be dynamic or static?
- c. What happened during the simulation? What was your size strategy? Please share your Python script with an explanation on how to use it.
- d. What are the different ways you can alter your trading strategy, so that you keep submitting orders but attempt to balance your book?
- e. Will market-making strategies typically work in a trending market?
- f. What was your P&L?

2.2 ALGO 2e - Group

- a. What was your strategy going in?
- b. What changes did you make to be able to handle multiple securities and fees/rebates?
- c. What happened during the simulation? What was your size strategy? Please share your Python script with an explanation on how to use it.
- d. What was your P&L?